

EcoSphere

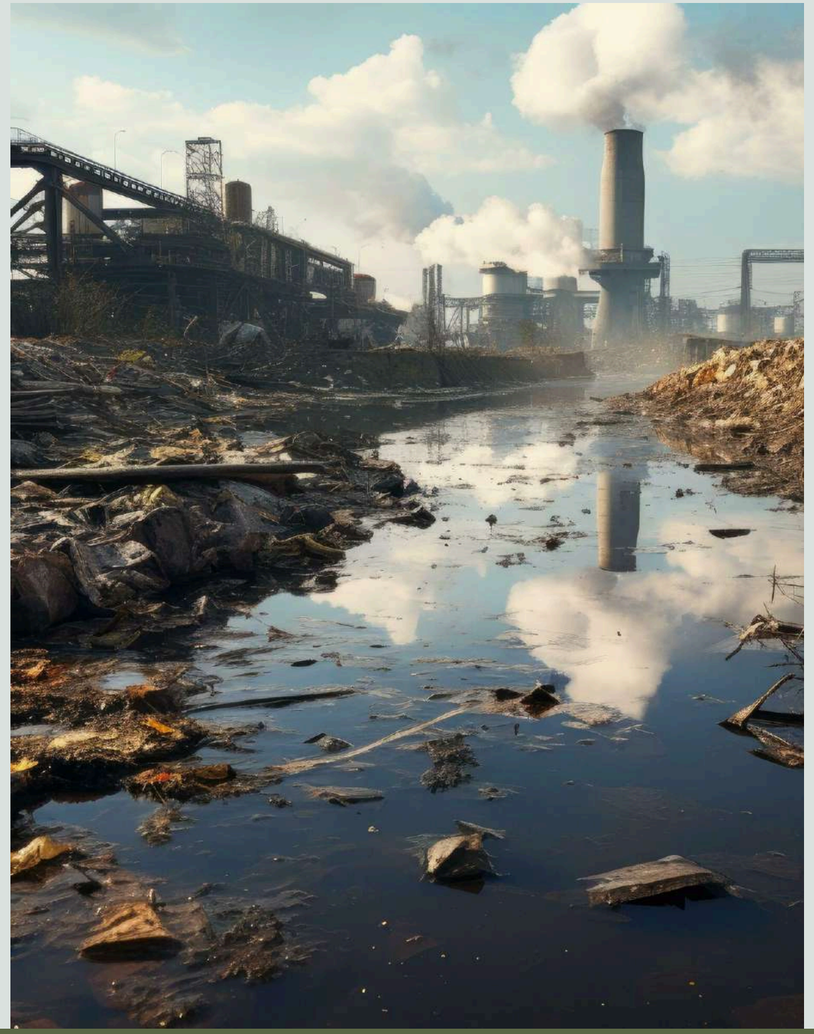
One Platform for a Sustainable Future

Team Name:- TechNova
Team Leader:- Nishtha Pandya
Team Members:- Aman Saini, Priyanshi

What we are solving???

The Sustainability Challenge

Despite growing environmental awareness, individuals and communities struggle to adopt sustainable practices because environmental activities are managed through multiple disconnected platforms. This fragmentation makes it difficult to track personal impact, stay motivated, and contribute consistently toward a greener future.



Key CHALLENGES

Waste Management

- Improper waste segregation due to lack of awareness.
- Limited access to recycling information.

Water Conservation

- Difficulty tracking daily water consumption.
- Water wastage often goes unnoticed.

Energy Consumption

- Users lack insights into electricity usage.
- No personalized recommendations to reduce consumption.

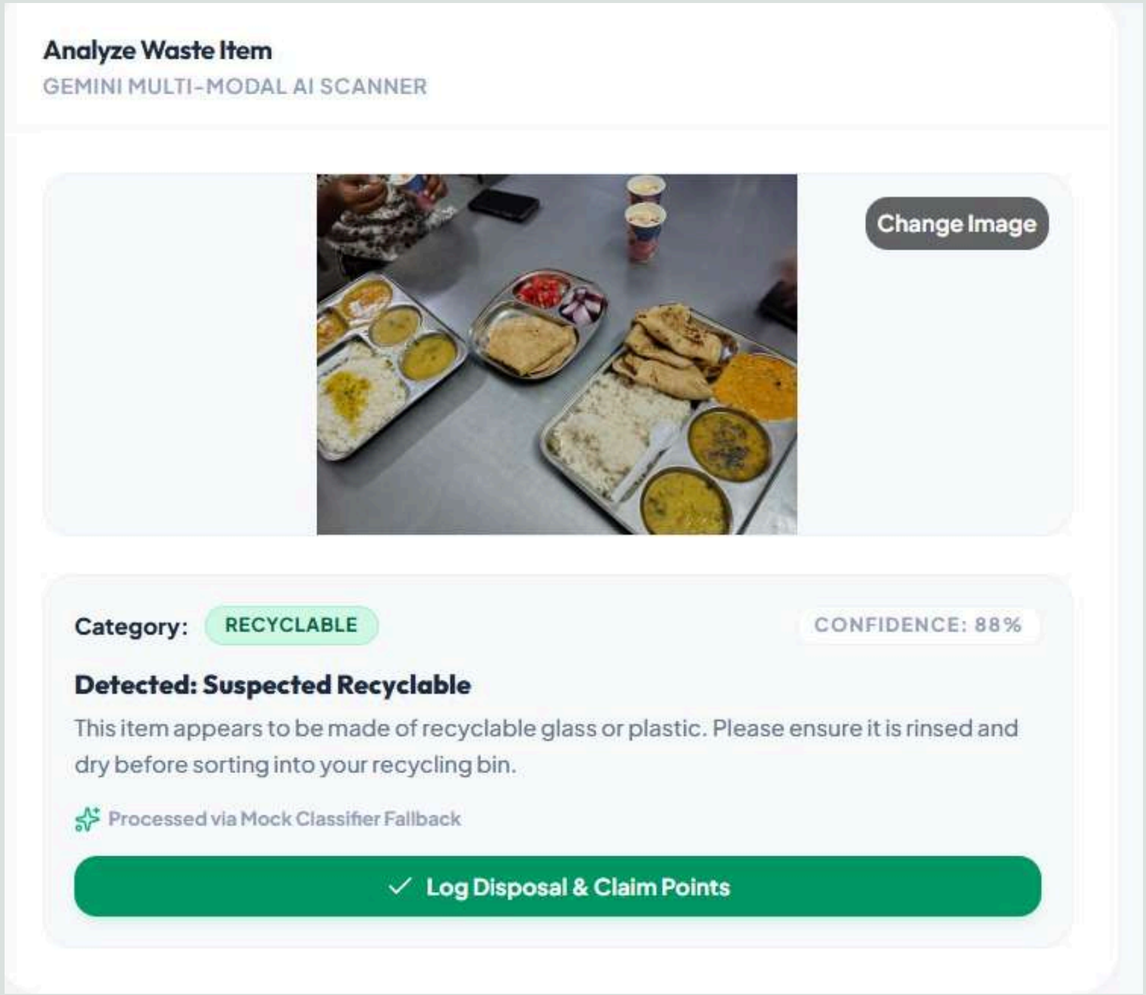
Carbon Footprint

- People are unaware of emissions generated through daily travel.
- Existing tools are complex and rarely used.

Community Engagement

- Sustainability efforts are mostly individual.
- No rewards, challenges, or collaborative motivation.

There is no single intelligent platform that enables users to monitor, manage, and improve all aspects of sustainable living in one place.



What's the solution?

EcoSphere is an AI-powered sustainability platform that unifies environmental management into a single, intelligent ecosystem. Instead of relying on multiple disconnected applications, users can monitor, improve, and measure their environmental impact through one integrated platform.
By combining AI, real-time analytics, gamification, and community engagement, EcoSphere transforms sustainable living into a simple, rewarding, and measurable experience.

How EcoSphere Solves the Problem?????

- Smart Waste Management – AI-based waste segregation & recycling rewards
- Water Conservation – Water usage tracking & saving recommendations
- Energy Monitoring – Electricity insights & AI-powered optimization
- Green Transportation – Carbon footprint calculation & eco-route suggestions
- Tree Plantation Tracker – Monitor plantations & community participation
- AI Sustainability Assistant – Personalized guidance & daily eco-challenges
- Sustainability Score – Unified score to measure overall environmental impact

EcoConnect

AI & IOT POWERED SMART GREEN CITY PLATFORM

What Makes EcoSphere Different?

Unified Sustainability Ecosystem
Brings waste, water, energy, transportation, and tree plantation into one intelligent platform.

AI-Driven Decision Support
Uses AI for waste classification, personalized recommendations, and sustainability assistance.

Unified Sustainability Score
Measures a user's overall environmental impact through a single, easy-to-understand green score.

Gamification & Community Engagement
Leaderboards, rewards, and eco-challenges motivate users to build lasting sustainable habits.

Actionable Insights
Transforms environmental data into personalized recommendations that help users make smarter, greener decisions.



Why EcoSphere is Feasible



Built with Proven Technologies

Developed using modern, industry-standard frameworks that ensure reliability and easy maintenance.

Modular Architecture

Each module (Waste, Water, Energy, Transportation, etc.) works independently, making the platform easy to develop and expand.

Cloud-Ready Deployment

Can be deployed on cloud platforms for seamless access, updates, and scalability.

Scalable Solution

Designed to grow from individual users to colleges, residential communities, organizations, and smart cities.

Future Integration Ready

Architecture supports integration with IoT devices, smart meters, and external APIs for real-time monitoring.

EcoSphere is designed as a practical, scalable, and real-world solution that can evolve from a hackathon prototype into a production-ready sustainability platform.



Environmental Impact

- Promote responsible waste segregation and recycling
- Encourage efficient water conservation
- Reduce unnecessary energy consumption
- Lower carbon emissions through sustainable transportation
- Increase tree plantation and green cover

Social Impact

- Build awareness about sustainable living
- Encourage community participation through challenges and leaderboards
- Develop long-term eco-friendly habits
- Empower individuals to make data-driven environmental decisions

Potential Outcomes (Demo Metrics)

♻️ 120 kg Waste Recycled
💧 2,000 L Water Conserved
🌳 50 Trees Registered
🌍 300 kg CO_2 Emissions Reduced
👥 250+ Active Users

Creating a Greener Future Through Technology

Supporting Global Sustainability

EcoSphere contributes to the United Nations Sustainable Development Goals by supporting:

- SDG 6 – Clean Water & Sanitation
- SDG 7 – Affordable & Clean Energy
- SDG 11 – Sustainable Cities & Communities
- SDG 12 – Responsible Consumption & Production
- SDG 13 – Climate Action
- SDG 15 – Life on Land

By transforming small daily actions into measurable environmental outcomes, EcoSphere empowers individuals and communities to collectively create a more sustainable future.



Technology Stack



FUTURE SCOPE

Barriers to Sustainable Growth

IoT Integration

Connect smart meters and sensors for real-time monitoring of water, energy, and waste.

Smart City Integration

Collaborate with municipalities and smart city initiatives for large-scale environmental monitoring.

Green Rewards Ecosystem

Introduce a carbon credit and reward system to encourage sustainable actions.

Blockchain-Based Transparency

Securely track green contributions and reward distribution using blockchain technology.

AI Predictive Analytics

Forecast resource consumption and provide proactive sustainability recommendations.

Wider Community Adoption

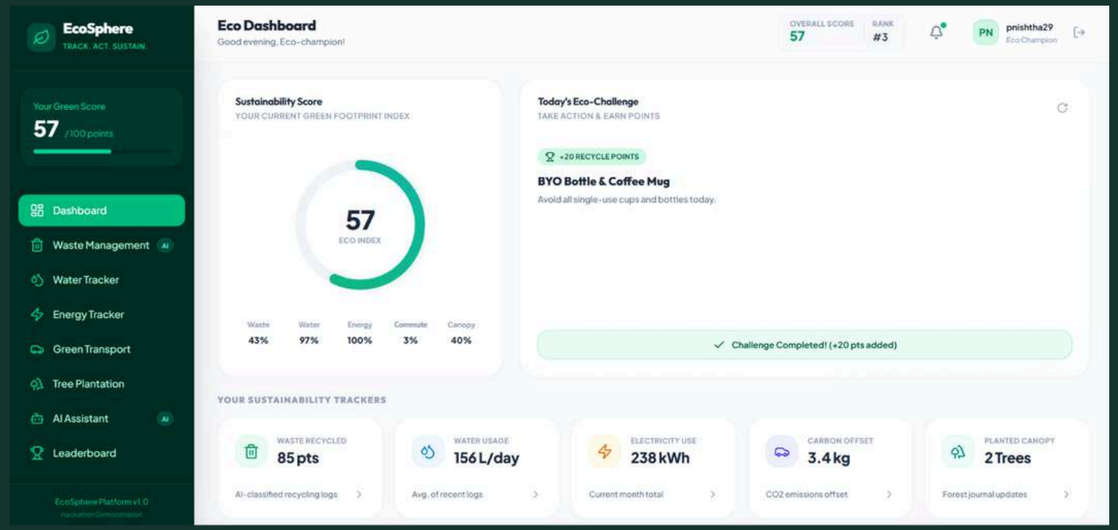
Deploy EcoSphere across colleges, residential societies, corporate campuses, and public communities.

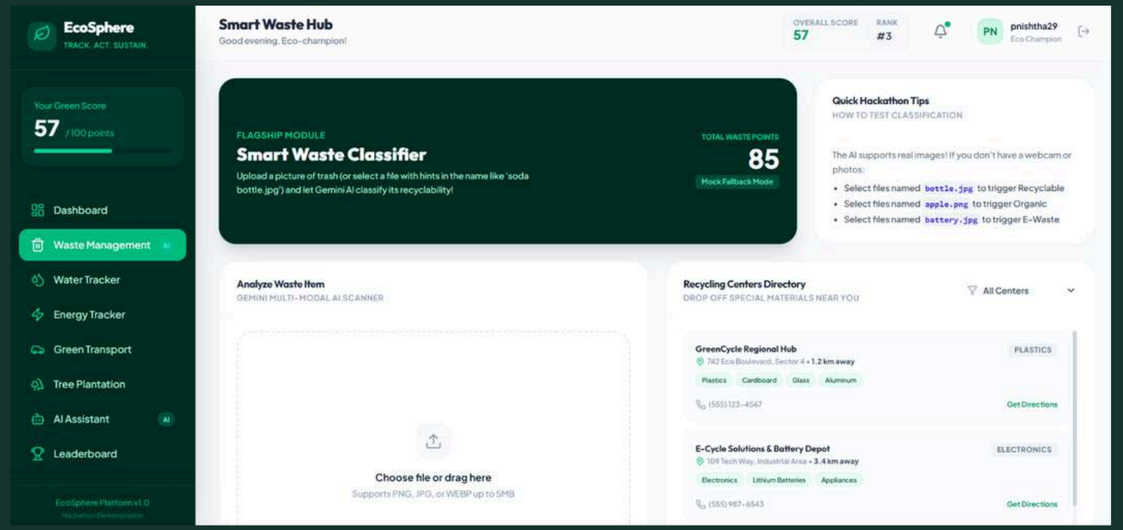
Our Vision:- To build a unified AI-powered sustainability ecosystem that empowers every individual, community, and city to make measurable contributions toward a greener future.

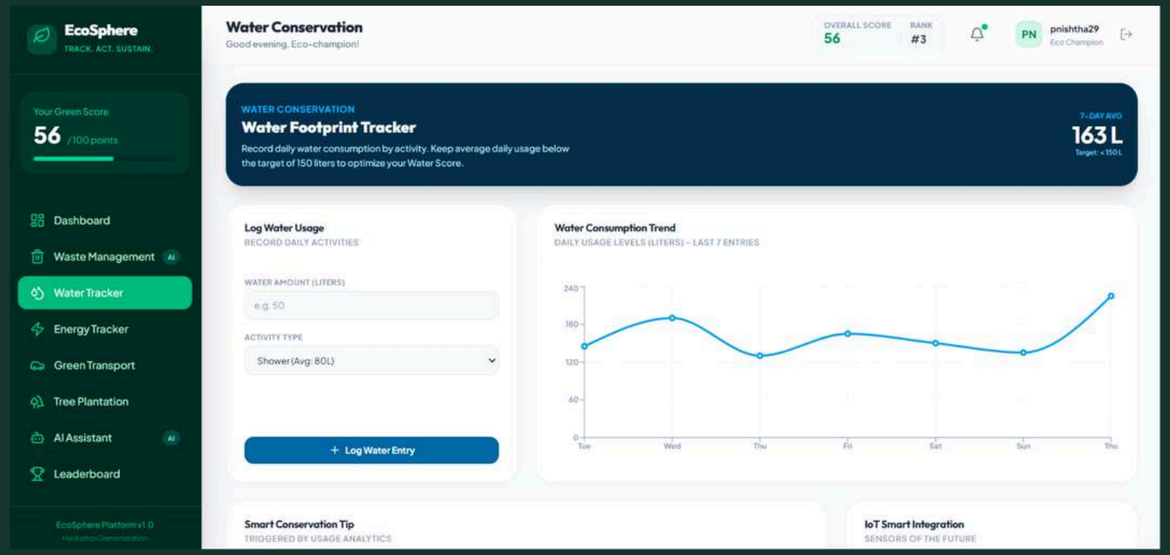
DEMO

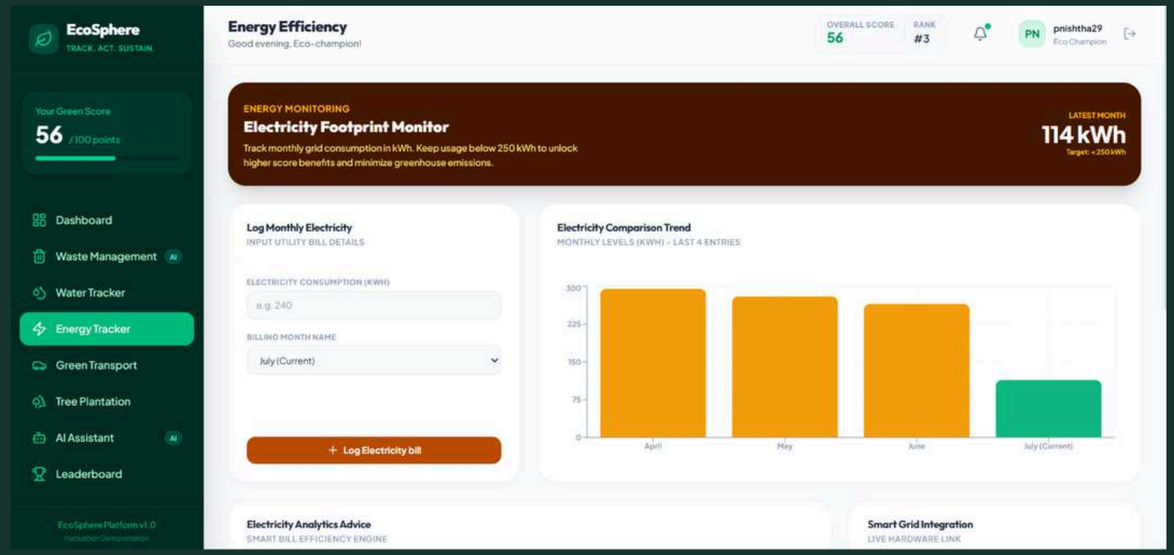
Project link:- https://eco-sphere-lac.vercel.app/

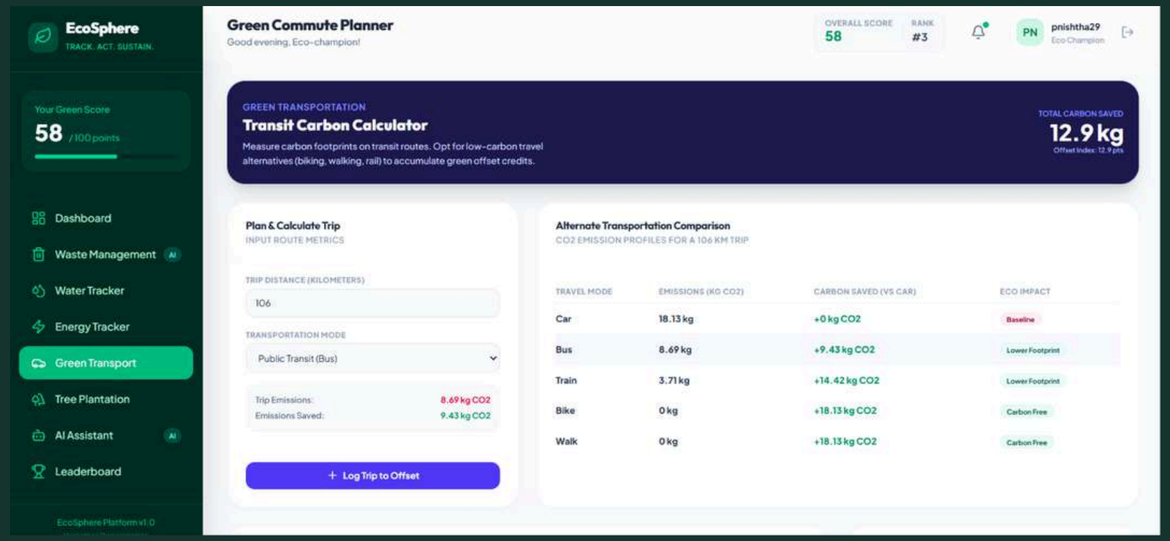
video link - https://drive.google.com/file/d/1tEP_IjutAqTsldmN9tShMioNSQJi5X0v/view?usp=sharing

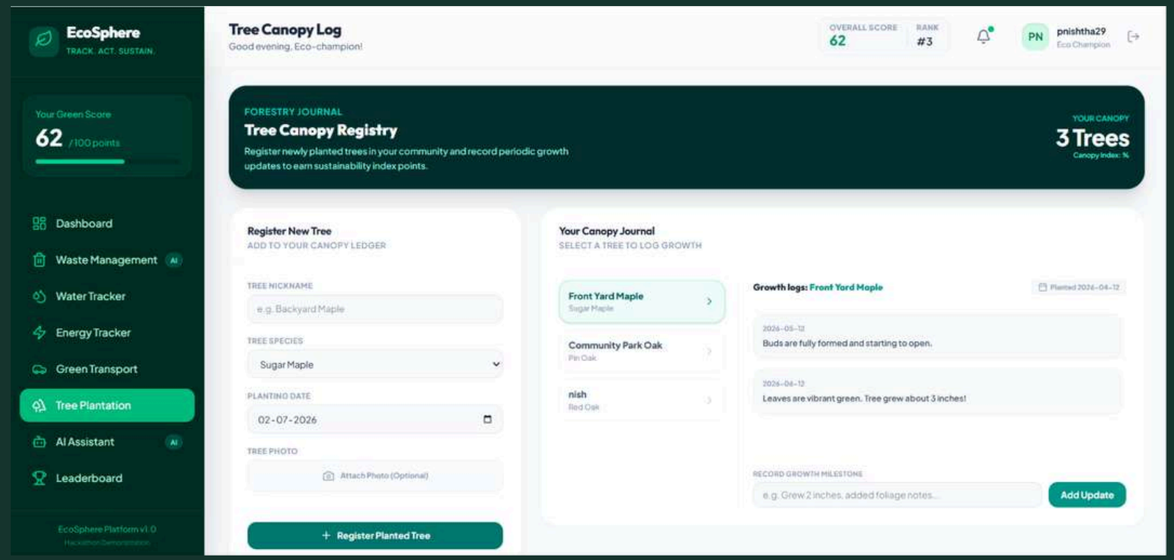





EcoSphere
TRACK. ACT. SUSTAIN.
Water Conservation
Good evening, Eco-champion!
OVERALL SCORE 56
RANK #3
PN pnishtha29 Eco Champion
Your Green Score
56 / 100 points
Dashboard
Waste Management AI
Water Tracker
Energy Tracker
Green Transport
Tree Plantation
AI Assistant AI
Leaderboard
EcoSphere Platform v1.0
Hackathon Demonstration
WATER CONSERVATION
Water Footprint Tracker
Record daily water consumption by activity. Keep average daily usage below the target of 150 liters to optimize your Water Score.
7-DAY AVG
163 L
Target: < 150 L
Log Water Usage
RECORD DAILY ACTIVITIES
WATER AMOUNT (LITERS)
e.g. 50
ACTIVITY TYPE
Shower (Avg: 80L)
+ Log Water Entry
Water Consumption Trend
DAILY USAGE LEVELS (LITERS) – LAST 7 ENTRIES
240
180
120
60
0
Tue
Wed
Thu
Fri
Sat
Sun
Thu
Smart Conservation Tip
TRIGGERED BY USAGE ANALYTICS
IoT Smart Integration
SENSORS OF THE FUTURE


EcoSphere
TRACK. ACT. SUSTAIN.
Energy Efficiency
Good evening, Eco-champion!
OVERALL SCORE 56
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Your Green Score
56 / 100 points
Dashboard
Waste Management AI
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Energy Tracker
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Leaderboard
EcoSphere Platform v1.0
Hackathon Demonstration
ENERGY MONITORING
Electricity Footprint Monitor
Track monthly grid consumption in kWh. Keep usage below 250 kWh to unlock higher score benefits and minimize greenhouse emissions.
LATEST MONTH
114 kWh
Target: < 250 kWh
Log Monthly Electricity
INPUT UTILITY BILL DETAILS
ELECTRICITY CONSUMPTION (KWH)
e.g. 240
BILLING MONTH NAME
July (Current)
+ Log Electricity bill
Electricity Comparison Trend
MONTHLY LEVELS (KWH) – LAST 4 ENTRIES
300
225
150
75
0
April
May
June
July (Current)
Electricity Analytics Advice
SMART BILL EFFICIENCY ENGINE
Smart Grid Integration
LIVE HARDWARE LINK


EcoSphere
TRACK. ACT. SUSTAIN.
Green Commute Planner
Good evening, Eco-champion!
OVERALL SCORE 58
RANK #3
PN pnishtha29 Eco Champion
Your Green Score
58 / 100 points
Dashboard
Waste Management AI
Water Tracker
Energy Tracker
Green Transport
Tree Plantation
AI Assistant AI
Leaderboard
EcoSphere Platform v1.0
GREEN TRANSPORTATION
Transit Carbon Calculator
Measure carbon footprints on transit routes. Opt for low-carbon travel alternatives (biking, walking, rail) to accumulate green offset credits.
TOTAL CARBON SAVED
12.9 kg
Offset Index: 12.9 pts
Plan & Calculate Trip
INPUT ROUTE METRICS
TRIP DISTANCE (KILOMETERS)
106
TRANSPORTATION MODE
Public Transit (Bus)
Trip Emissions: 8.69 kg CO2
Emissions Saved: 9.43 kg CO2
+ Log Trip to Offset
Alternate Transportation Comparison
CO2 EMISSION PROFILES FOR A 106 KM TRIP
TRAVEL MODE
EMISSIONS (KG CO2)
CARBON SAVED (VS CAR)
ECO IMPACT
Car 18.13 kg +0 kg CO2 Baseline
Bus 8.69 kg +9.43 kg CO2 Lower Footprint
Train 3.71 kg +14.42 kg CO2 Lower Footprint
Bike 0 kg +18.13 kg CO2 Carbon Free
Walk 0 kg +18.13 kg CO2 Carbon Free


EcoSphere
TRACK. ACT. SUSTAIN.
Tree Canopy Log
Good evening, Eco-champion!
OVERALL SCORE 62
RANK #3
PN pnishtha29 Eco Champion
Your Green Score
62 / 100 points
Dashboard
Waste Management AI
Water Tracker
Energy Tracker
Green Transport
Tree Plantation
AI Assistant AI
Leaderboard
EcoSphere Platform v1.0
Hackathon Demonstration
FORESTRY JOURNAL
Tree Canopy Registry
Register newly planted trees in your community and record periodic growth updates to earn sustainability index points.
YOUR CANOPY
3 Trees
Canopy Index: N
Register New Tree
ADD TO YOUR CANOPY LEDGER
TREE NICKNAME
e.g. Backyard Maple
TREE SPECIES
Sugar Maple
PLANTING DATE
02-07-2026
TREE PHOTO
Attach Photo (Optional)
+ Register Planted Tree
Your Canopy Journal
SELECT A TREE TO LOG GROWTH
Front Yard Maple
Sugar Maple
Community Park Oak
Pin Oak
nish
Red Oak
Growth logs: Front Yard Maple
Planted 2026-04-12
2026-05-12
Buds are fully formed and starting to open.
2026-04-12
Leaves are vibrant green. Tree grew about 3 inches!
RECORD GROWTH MILESTONE
e.g. Grew 2 inches, added foliage notes...
Add Update

Thank You!!!!!!!!!!